



Similar to IBM's autonomic computing is Hewlett-Packard's Planetary Computing. Researchers at HP's laboratories in Palo Alto, California, are designing a 50,000-node programmable data center that is physically wired once, but can be rewired programmatically. A data centre control system will determine what's available and then install, configure, deploy, monitor and assure services worldwide. This system will be self-monitoring, self-healing and self-adapting

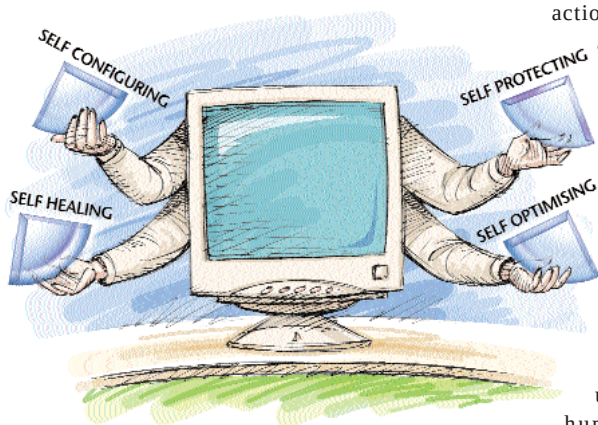
tain, diagnose and heal themselves.

Somebody has been listening hard. IBM, for instance, had launched an 'autonomic computing' initiative in its research division in 2001. Big Blue is already working on the next generation of computing which it calls 'autonomic' because it believes the computer must act like our autonomic nervous system.

The autonomic computing project is based on the concept of SMASH—Simple, Many and Self-Healing. This concept was applied in Deep Blue, the IBM computer that beat chess champion Garry Kasparov at chess a few years ago. IBM's present project, Blue Gene, an MPP (massively parallel processing) computer designed to map genomes, builds on Deep Blue but is larger and more sophisticated.

Key characteristics

Autonomic computing innovations are based on the four key characteristics of self-managing computing systems: self-configuring, self-healing, self-optimising and self-protecting. The architecture envisioned by IBM begins with the premise that implementing self-managing attributes involves an intelligent control loop. The autonomic manager is a component that implements the control loop. This



Autonomic Computing at Work

A student of the Archbishop Moeller High School, Cincinnati, walked up to the school IT administrator Keith Schaeper, with a corrupted operating system. The laptop was as good as useless. Nothing would run. Keith used Xpoint's software, and had a fully operating laptop in about three-and-a-half minutes. Corrupted operating systems, disabling viruses, and the like are now heartburns of the past for the students of this school. With nearly 1,000 laptops in use by students and faculty, the task of protecting and managing each laptop is daunting. Keith has deployed all the laptops with Xpoint's Managed Recovery Software for the client. Each laptop contains a generic image for that system, as well as a backup of the system with all personalised settings.

IBM has collaborated with Xpoint to add Rapid Restore to its PCs. Now, with just the push of a button (the [F11] key), IBM ThinkPads and NetVista desktops can automatically reimage themselves and restore user data. According to Xpoint, this PC is a managed recovery solution designed to protect the software image running on the client system, as well as user data files. The PC can restore the sys-

tem from the IBM Service Partition on the client system hard disk drive, often in as little as 15 to 20 minutes. Local recovery is made possible through an [F11] function key system recovery option that provides diskette-less managed recovery. The PC helps manage and quickly restore all client systems operating in the field, while enabling end users to manage their own backup and recovery.

The PC enables the user to have up to five levels of backup and recovery options. A key feature of this product is the ability of end users to recover from simple mistakes. For example, a user may have created a critical document, which was later corrupted or deleted by mistake. With this PC, the user can find the lost file (as long as it was created prior to the last backup) and restore the individual file using the Windows Explorer interface. After the backup and IBM Service Partition has been set up, protection is automatic. Backups will commence based on the default schedule, which can easily be adjusted to a customised time and frequency. In addition, an on-demand backup feature provides instantaneous backup when an important file must be protected.

loop collects information from the system, makes decisions and then adjusts the system as necessary. An intelligent control loop can enable the system to do such things as:

Self-configure: It's the ability to dynamically configure itself on the fly and with minimal human intervention. The loop will begin installing software when it detects that software is missing.

Self-heal: This is the ability to detect improper operations and initiate corrective action without disrupting system

applications. Corrective action could mean that a product alters its own state or influences changes in other elements. Day-to-day operations do not falter because of events at the component level.

Self-optimize: This is the ability to efficiently maximise resource allocation and utilisation to meet the users' needs, with minimal human intervention. In the near

term, self-optimisation addresses the complexity of managing system performance. **Self-protect:** The goal is to provide the right information to the right users at the right time through actions that grant access based on the users' role and pre-established policies. It can detect hostile or intrusive behaviour as it occurs and take autonomous actions to make itself less vulnerable to unauthorised access and use, viruses, denial-of-service attacks and general failures. Self-protection is also about recognising and dealing with overload conditions that could jeopardise the integrity of the system.

Is it already here?

Some steps toward autonomic computing have already been taken. Take the case of Symantec with their Live Update feature. The company needed to rapidly update its customers' data and software to recognise the latest viruses. So it established a database from which all users could do a 'live update' of their system, thus ensuring that each customer would have the latest virus protection. "Live update decreases the